

Designing an Academic and Career Development Website for Computer Science Students

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Team 02

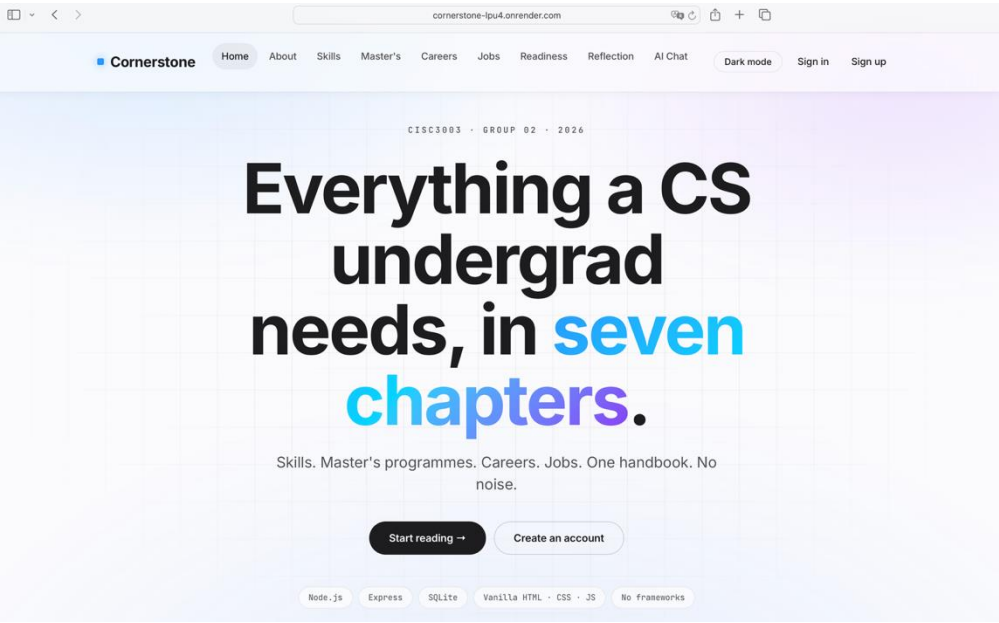


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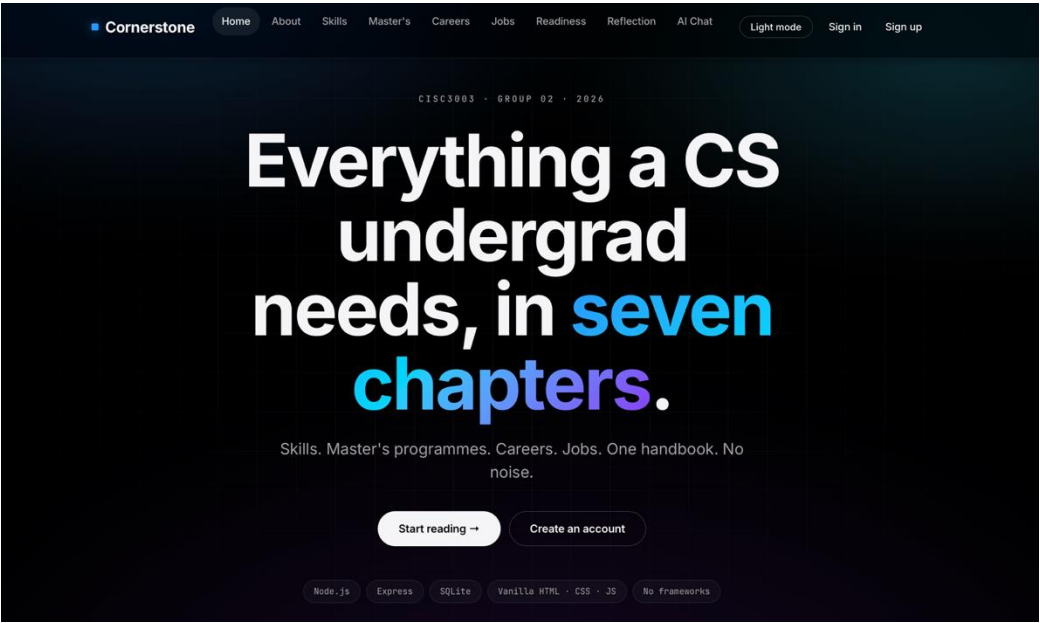
My contribution in the Team project

1. Overall Page design
2. Design and develop Master's Page
3. Design and develop Career's Page
4. Design and develop Job's page
5. Design and develop AI Chat page

Overall Page design

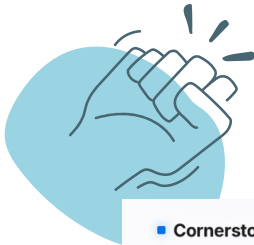
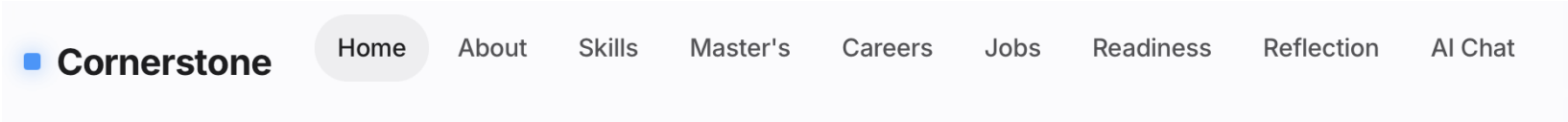


Light mode

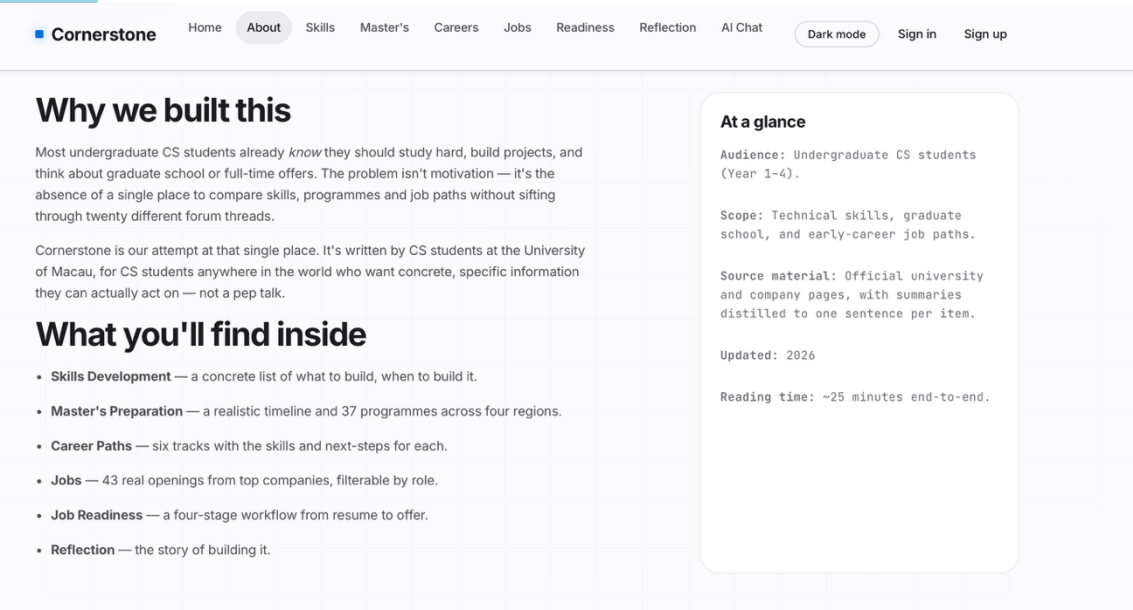


Dark mode

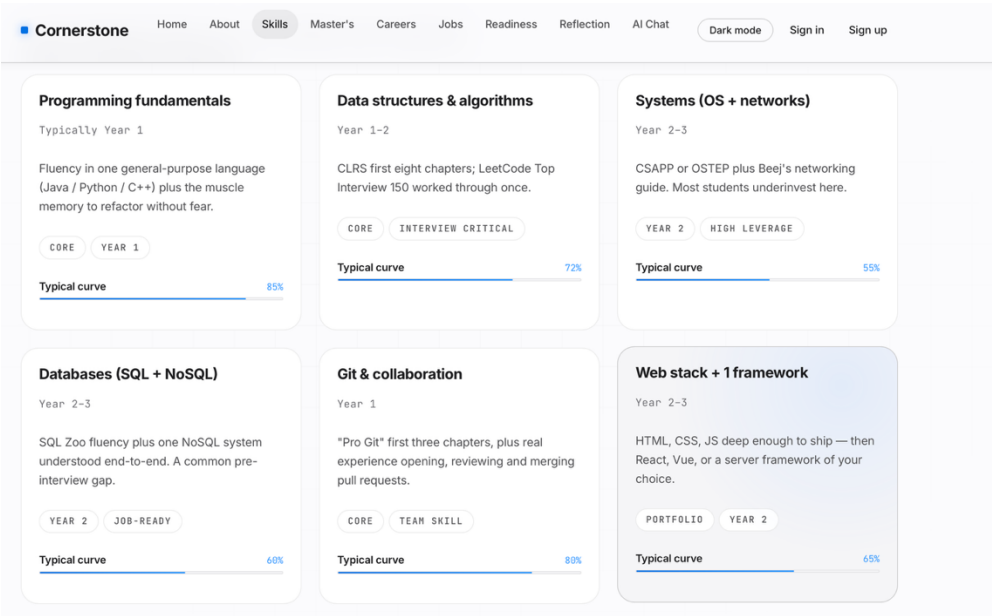
My main contribution 1 - Overall Page design



The website is organized into eight sections.

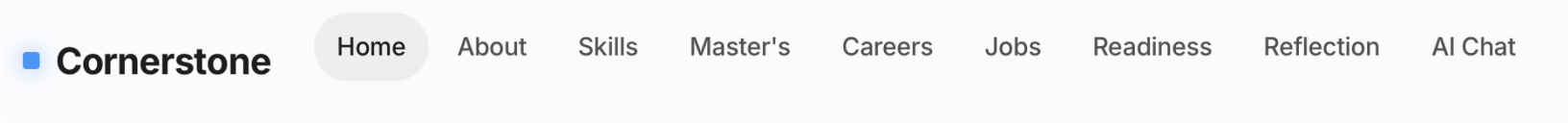


About section

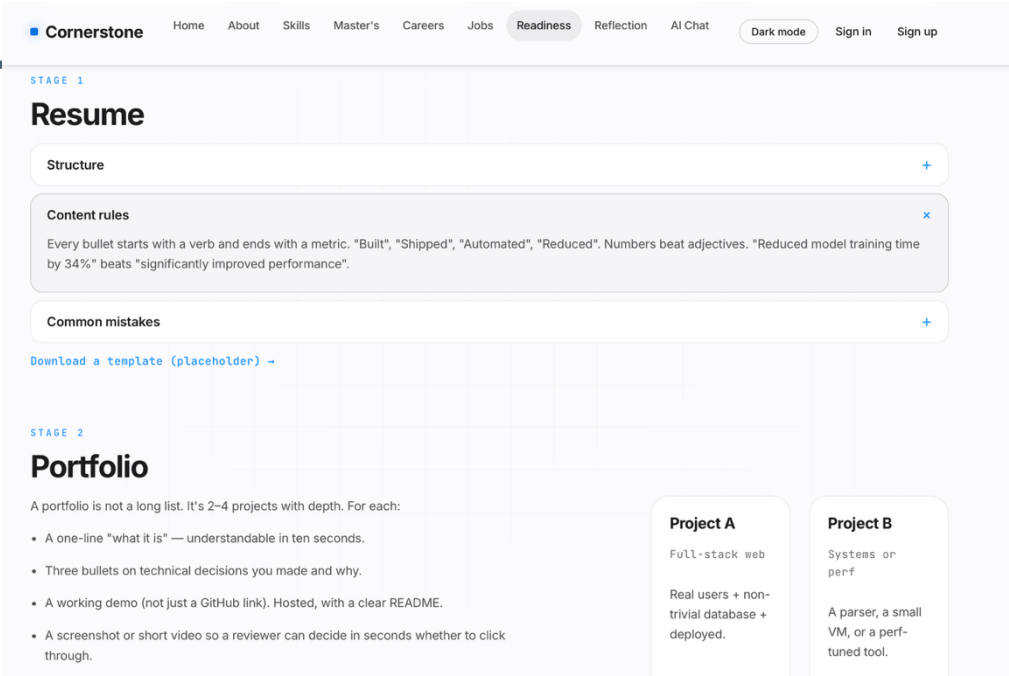


Skills section

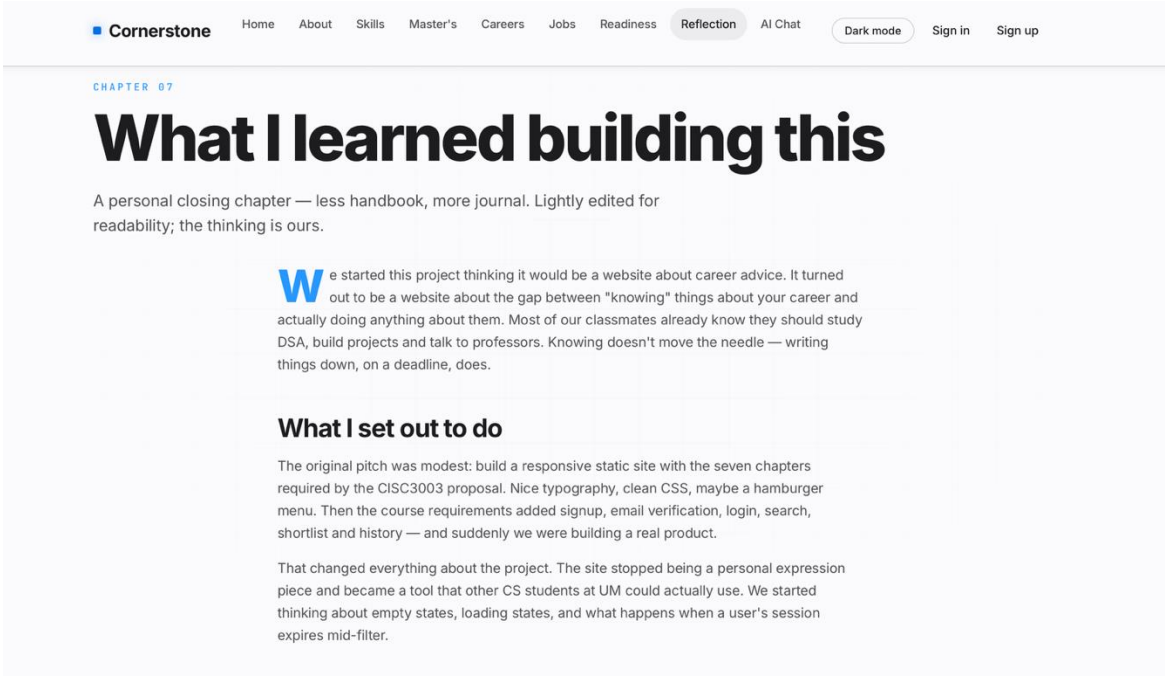
My main contribution 1 - Overall Page design



The website is organized into eight sections.



Readiness section



Reflection section

My main contribution 2 – Design and develop Master’s page



■ Cornerstone

Home

About

Skills

Master's

Careers

Jobs

Readiness

Reflection

AI Chat

Dark mode

Sign in

Sign up

CHAPTER 03

Master's preparation

A realistic, eighteen-month timeline, honest advice on the essays and tests, and 37 programmes organised by region and tier — so you can compare like with like.

≥3.3
TYPICAL GPA TARGET

2+
RECOMMENDATION LETTERS

6–12
MONTHS OF LEAD TIME

1
CLEAR RESEARCH INTEREST

ROADMAP

1

YEAR 2 · EXPLORE

Find the intersection of "interesting" and "good at"

- Take one research-heavy elective you'd enjoy reading about on a Saturday.
- Read three abstracts from conferences in that subfield (NeurIPS, SIGCOMM, CHI — depending on area).
- Approach a professor about a small research task; even volunteer work counts.

First, on the Master’s page, I introduced how students can prepare for graduate school applications.

2

YEAR 3 FALL · SHORTLIST

Draft a programme list by region and tier

- Pick 3 reach, 4 target and 3 safety programmes (total 10).
- For each, note: faculty whose work you'd want to join, deadlines, required tests, tuition.
- Start following those labs on X, GitHub or group pages.

LOGISTICS

3

YEAR 3 SPRING · RESEARCH + TESTS

Produce one artefact and book one exam

- Aim for a small research contribution (code, dataset, or poster) you can point to.
- Schedule TOEFL/IELTS first, GRE only if required by target programmes.
- Start a spreadsheet of every programme's deadline and required document.

ACADEMIC

LOGISTICS

4

SUMMER BEFORE YEAR 4 · ESSAYS + LETTERS

Draft your Statement of Purpose and ask for letters

- Write a SoP that connects what you've done → why this programme → what you'll do next.
- Ask recommenders 8+ weeks before the first deadline. Give them a bullet list of context.
- Polish CV to fit the genre expected by graduate committees (not industry resumes).

WRITING

HIGH LEVERAGE

My main contribution 2 – Design and develop Master’s page



DEEP DIVES

Essays, letters, tests and research

Statement of Purpose +

Recommendation letters +

CV for graduate applications x

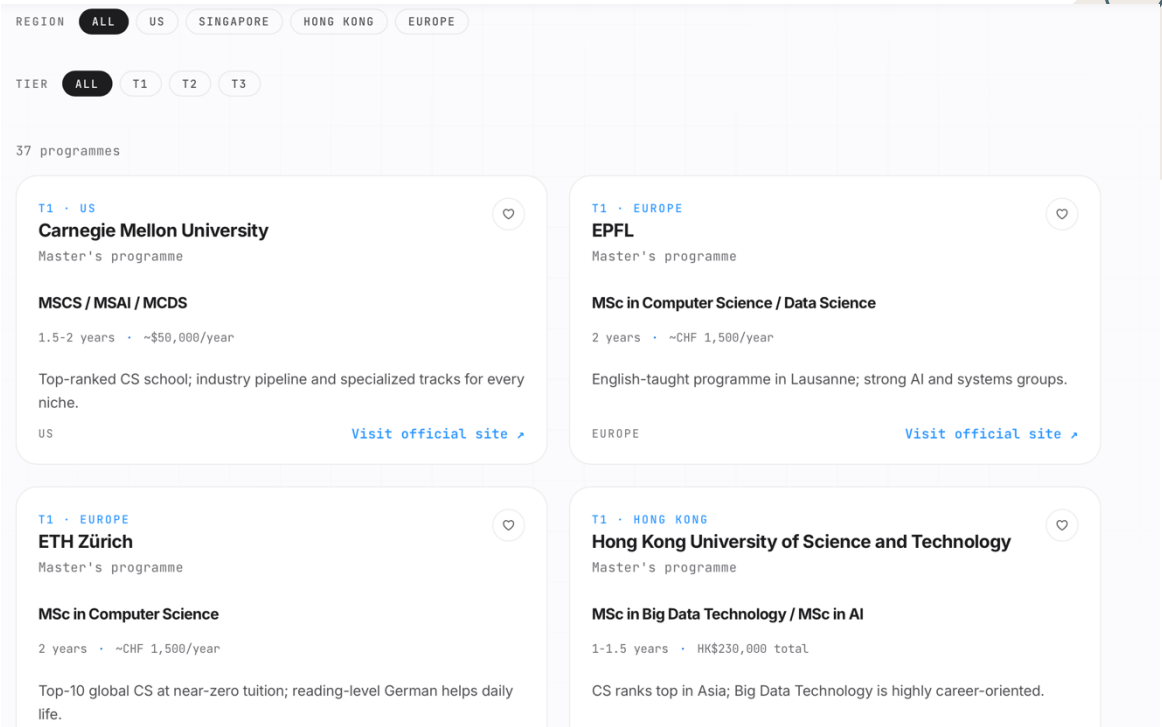
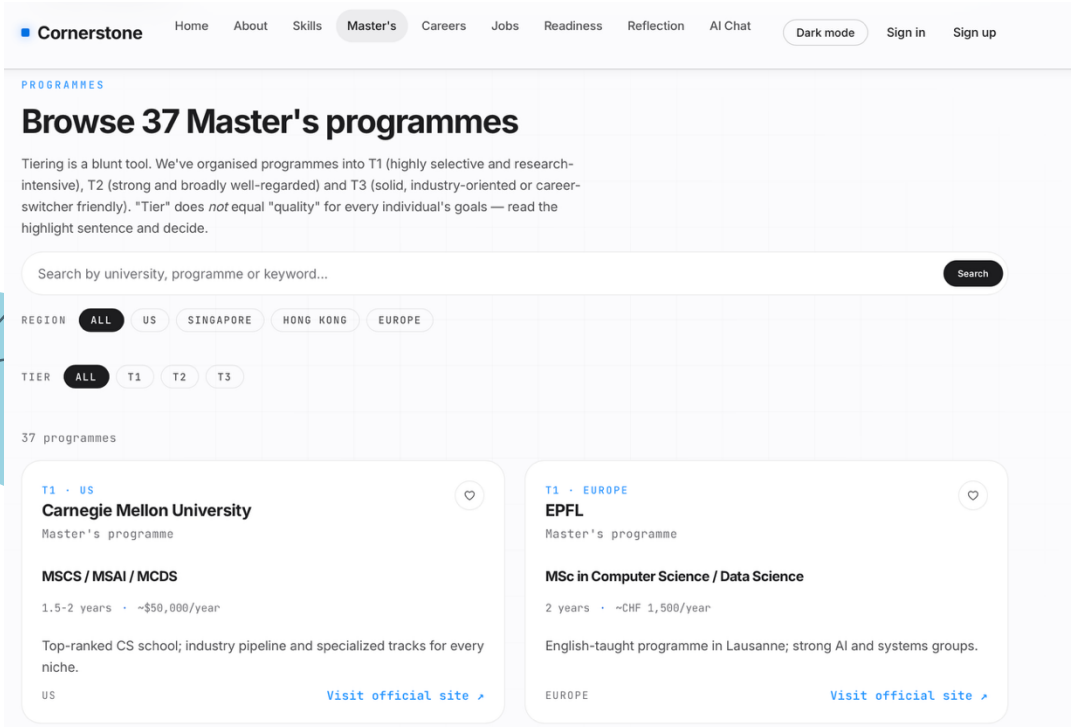
Graduate CVs look different from industry resumes. Lead with education, then research experience, then publications or projects, then teaching, then industry. Expected length: 2 pages, never 1. Include a skills section but don't rely on it.

Standardized tests +

Research experience +

Second, I introduced the materials required for Master’s applications and explained how to prepare them effectively.

My main contribution 2 – Design and develop Master’s page



I introduced a wide range of graduate programs to help students gain a better understanding of their options.

My main contribution 2 – Design and develop Master’s page



Tiering is a blunt tool. We've organised programmes into T1 (highly selective and research-intensive), T2 (strong and broadly well-regarded) and T3 (solid, industry-oriented or career-switcher friendly). "Tier" does *not* equal "quality" for every individual's goals — read the highlight sentence and decide.

Search by university, programme or keyword...

Search

REGION

ALL

US

SINGAPORE

HONG KONG

EUROPE

TIER

ALL

T1

T2

T3

Users can search by keywords, or filter programs based on region and level of application difficulty.

My main contribution 2 – Design and develop Master’s page



17 programmes

T1 · US

Carnegie Mellon University

Master's programme

MSCS / MSAI / MCDS

1.5-2 years · ~\$50,000/year

Top-ranked CS school; industry pipeline and specialized tracks for every niche.

US

Visit official site

T1 · US

Massachusetts Institute of Technology

Master's programme

EECS Master of Engineering (MEng)

1-2 years · ~\$60,000/year

Research-heavy programme with strong AI and systems groups; MEng open to MIT undergrads.

US

Visit official site

T1 · US

Princeton University

Master's programme

MSE in Computer Science

2 years · ~\$60,000/year

Small, research-focused; often a feeder into Princeton's PhD programme.

US

Visit official site

T1 · US

Stanford University

Master's programme

MS in Computer Science

1.5-2 years · ~\$62,000/year

Specialization tracks in AI, HCI, Systems; Silicon Valley network is unmatched.

US

Visit official site

We also provide direct links: by clicking the buttons, users can navigate to the corresponding graduate programs at each university for more detailed information.

MIT EECS

Electrical Engineering Computer Science Artificial Intelligence + Decision-making

ACADEMICS RESEARCH PEOPLE COMMUNITY & BELONGING NEWS & EVENTS ABOUT

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MIT

ACADEMICS

Graduate Programs

ACADEMICS

Past Terms' Subject Updates and WTW

Subject Updates

Subject numbering

Undergraduate programs

Graduate Programs

IAP offerings

My main contribution 2 – Design and develop Careers page



Cornerstone

HomeAboutSkillsMaster'sCareersJobsReadinessReflectionAI ChatDark modeSign inSign up

CHAPTER 04

Career paths

Nine possible directions after a CS undergrad — what the daily work looks like, what skills matter, and where to go next.

FILTER BY AREA

ALLSOFTWAREAI / MLDATASECURITYRESEARCHPRODUCT

Showing all 9 career paths.

Software engineering

The default path. Builds and maintains the systems that everyone else uses. Roles span web, mobile, backend infra, DevOps, and platform.

DSASYSTEMSGIT

Typical roles: SWE, Full-stack, Backend, SRE, Platform.

See SDE openings →

Machine learning engineering

Builds production ML systems — training pipelines, inference infrastructure, model evaluation. Requires strong ML fundamentals and software engineering chops.

PYTORCHDSASTATISTICS

Typical roles: MLE, Applied Scientist, ML Platform Engineer.

See MLE openings →

Data science & analytics

Turns product and business questions into measurable experiments. SQL-heavy with a healthy dose of causal inference and experimental design.

SQLSTATISTICSPYTHON

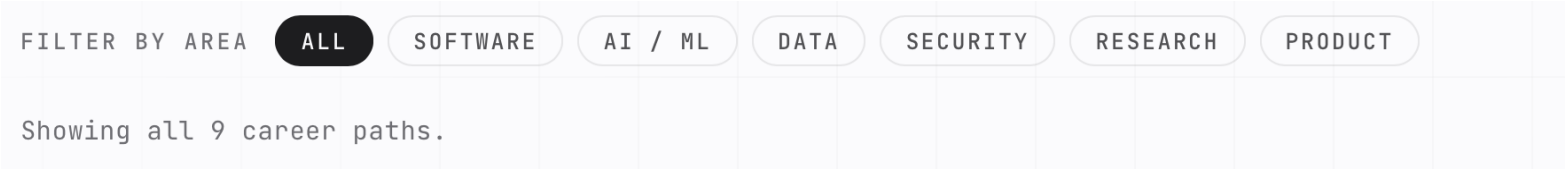
Typical roles: Data Scientist, Analytics Engineer, Product DS.

See Data openings →

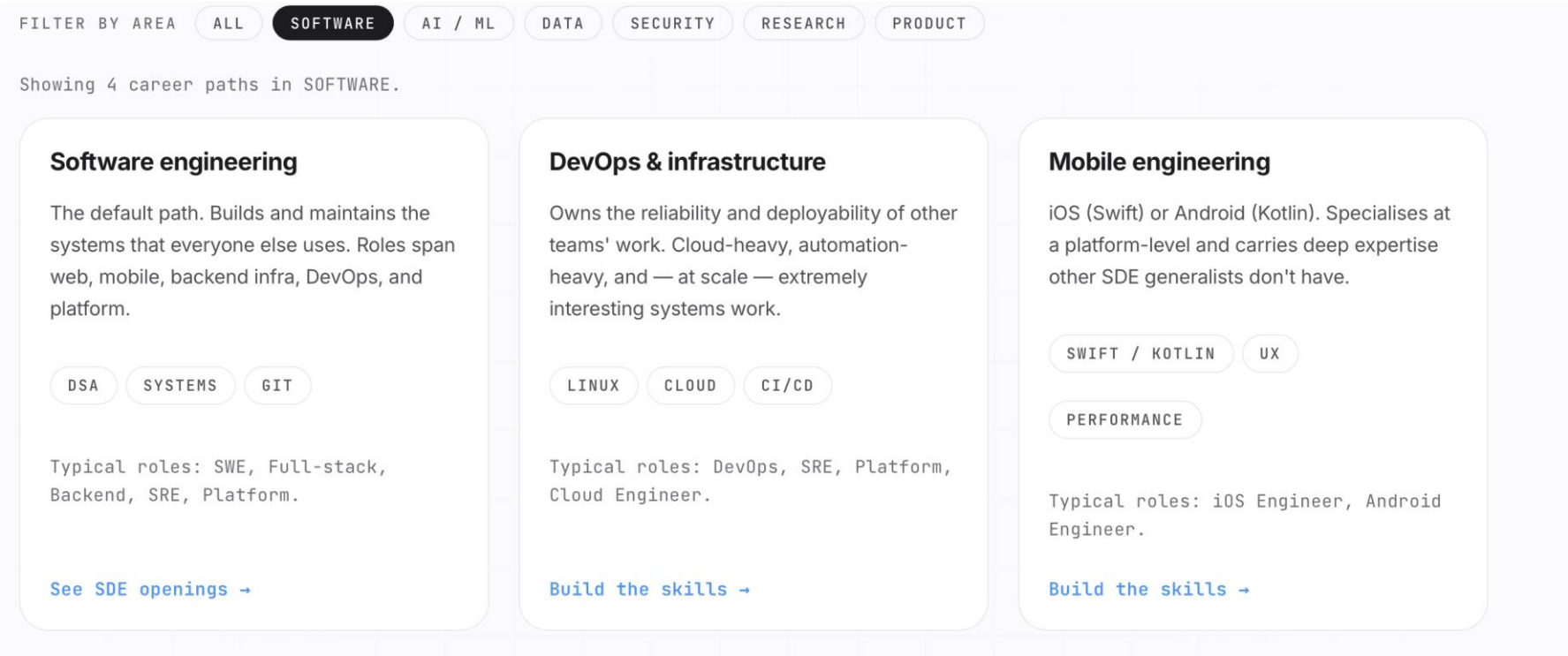
To help students gain a more detailed understanding of career planning, we designed a Careers page.

My main contribution 2 – Design and develop Careers page

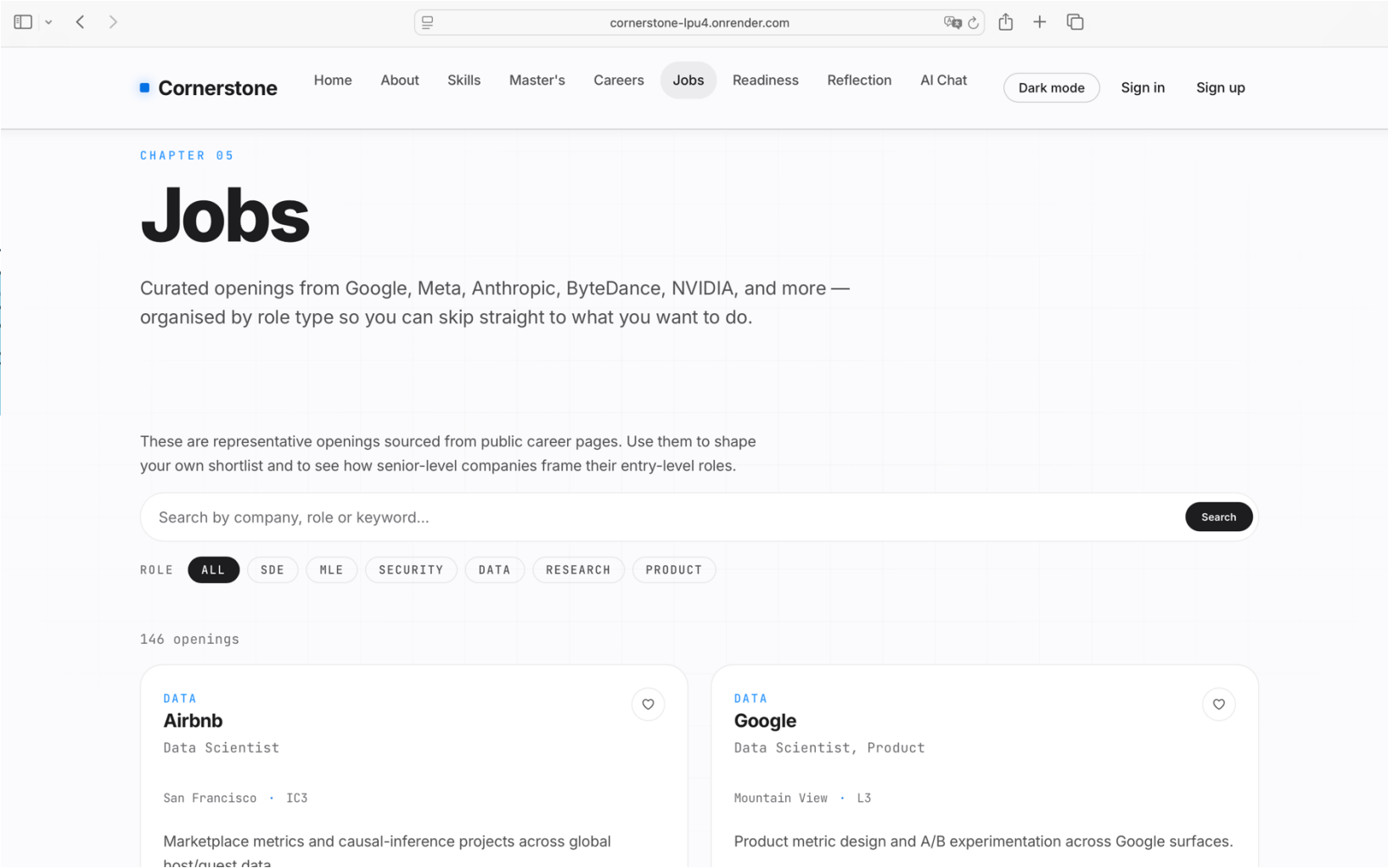
We provide six career categories



By clicking on a specific category, users can view the detailed types of jobs included within it.



My main contribution 4 – Design and develop **Jobs** page



The main purpose of the Jobs page is to help students submit their resumes and understand which positions are currently open for applications.

My main contribution 4 – Design and develop Jobs page



These are representative openings sourced from public career pages. Use them to shape your own shortlist and to see how senior-level companies frame their entry-level roles.

Search

ROLE

ALL

SDE

MLE

SECURITY

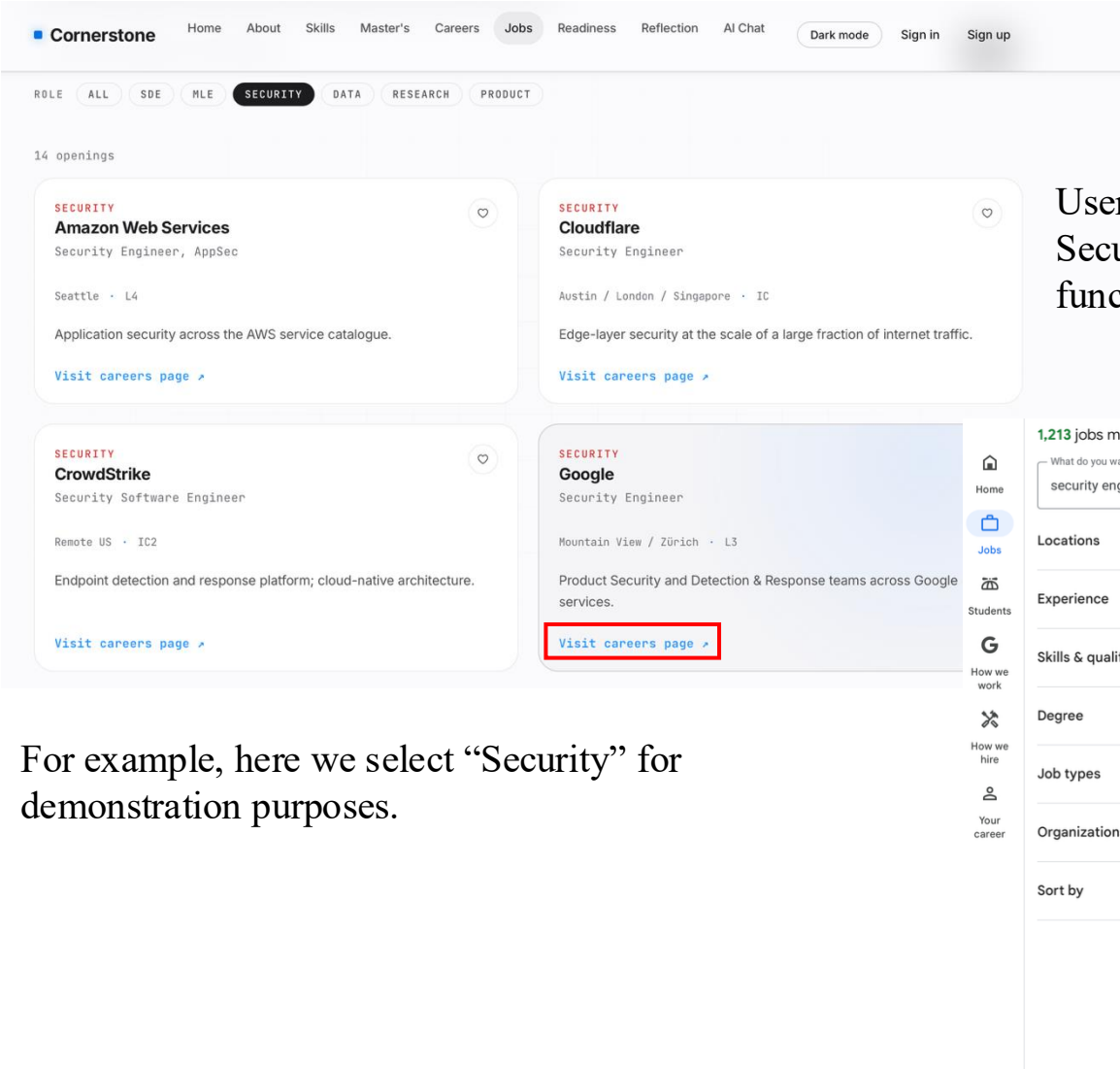
DATA

RESEARCH

PRODUCT

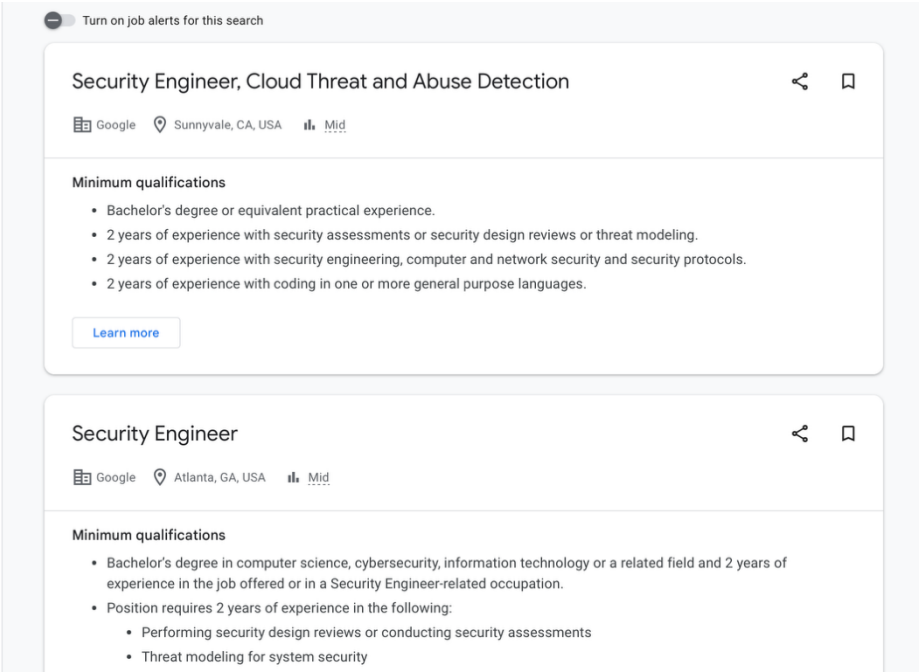
We divided all positions into six categories, allowing users to search for relevant jobs either by category filters or by keywords.

My main contribution 4 – Design and develop **Jobs** page

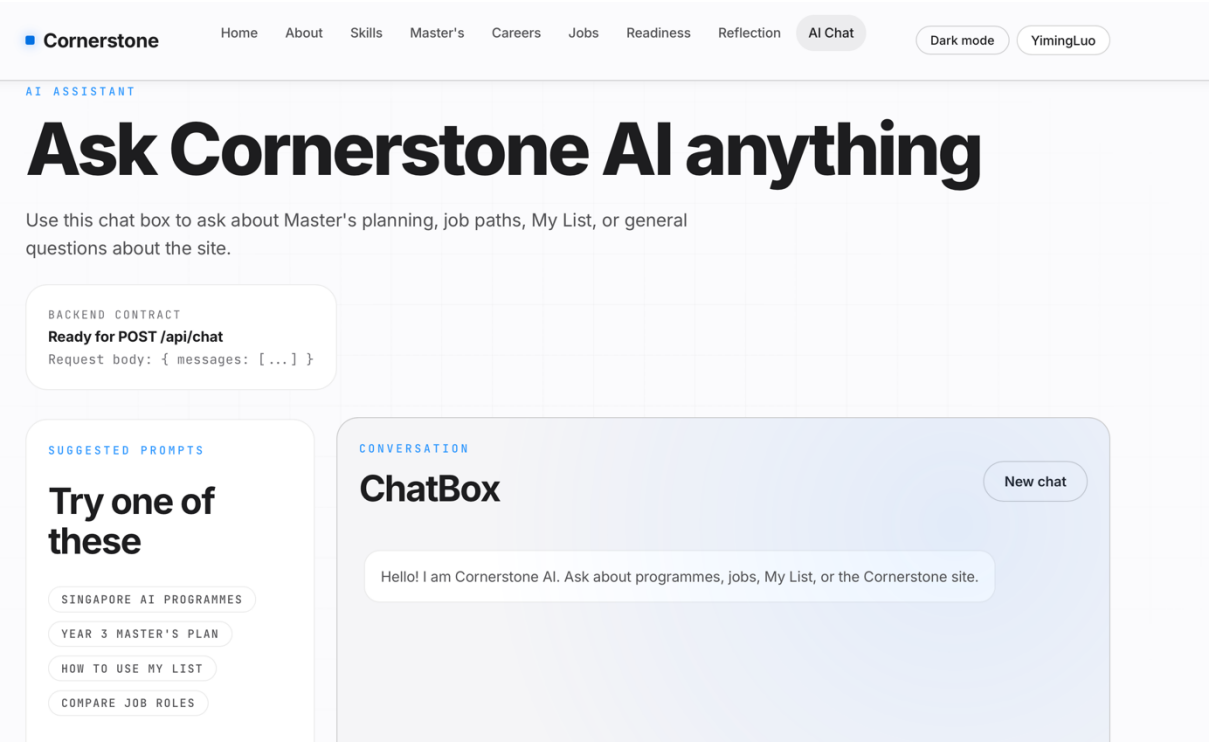


Users can then click on a relevant company, which will redirect them to all Security-related positions offered by that company. In this example, this function is implemented for Google.

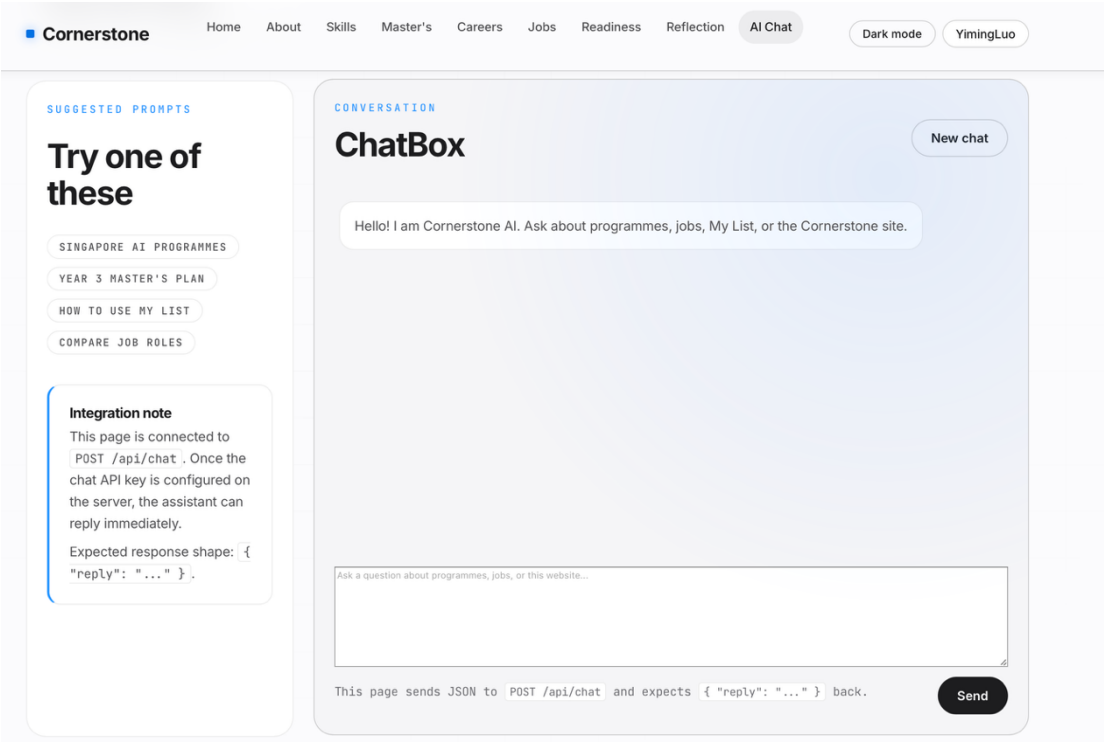
For example, here we select “Security” for demonstration purposes.



My main contribution 5 – Design and develop AI Chat page

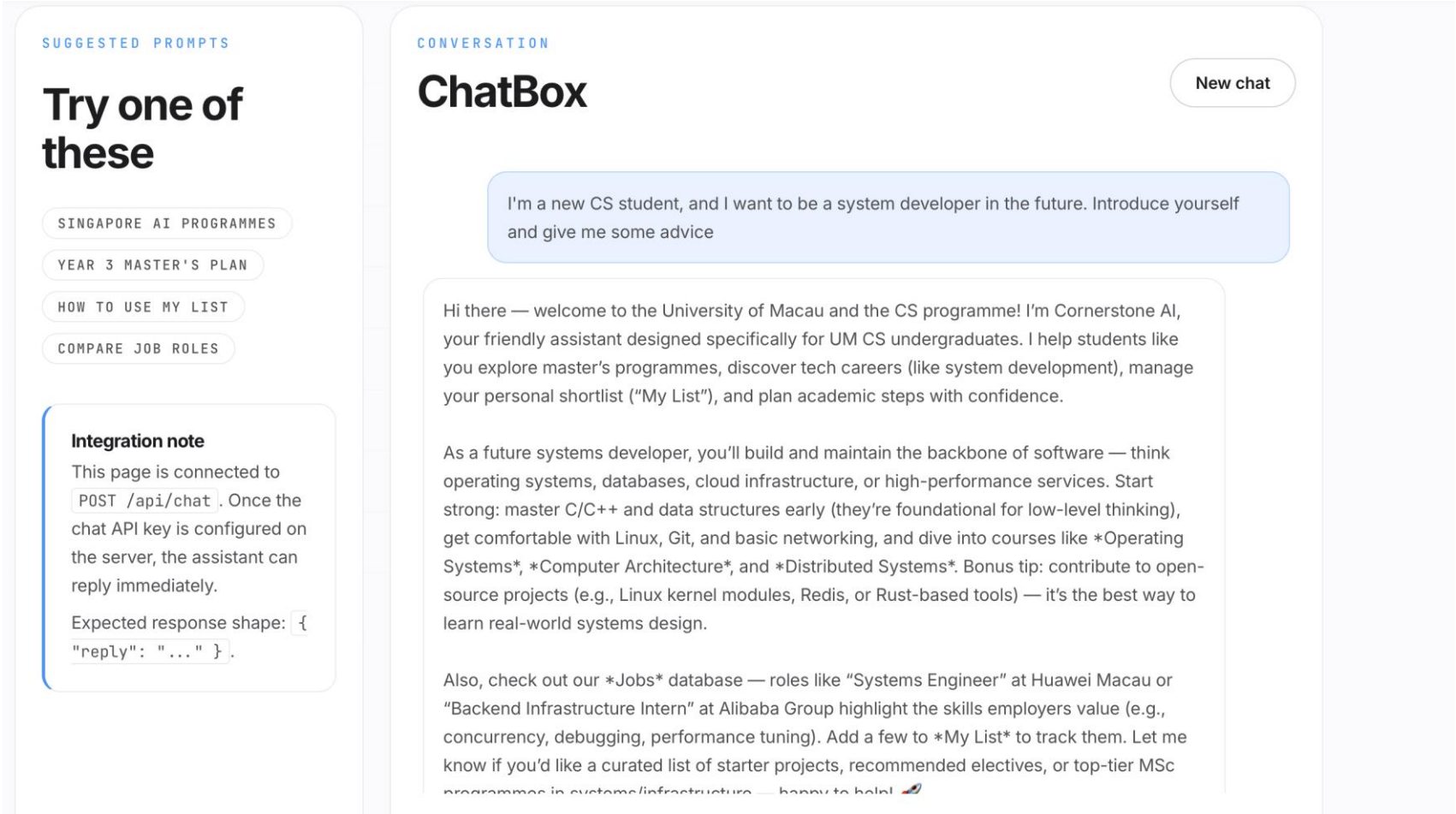


We also provide an AI assistant to help users with career and application planning.



After logging in, users can interact with the AI assistant and ask it to help them with personalized planning.

My main contribution 5 – Design and develop AI Chat page



The AI Chat feature helps users make better use of the website and provides additional career advice based on the website’s content.

Summary

Designed and developed the overall page structure

Designed the general layout and structure of the website.

Organized the website into multiple sections.

Helped users navigate academic, career, job, and AI-assisted planning resources.

Designed and developed the Master's Page

Introduced how students can prepare for Master's applications.

Presented a wide range of graduate programs.

Added keyword search and filters based on region and application difficulty.

Provided direct links to official university graduate program pages.

Designed and developed the Careers Page

Designed the page to support students' career planning.

Helped students understand different career directions in computer science.

Designed and developed the Jobs Page

Designed the page to help students find open positions.

Helped students understand which jobs are currently available for application.

Divided all positions into six categories, create category-based filtering and keyword-based job search.

Provided company links for specific job categories.

Designed and developed the AI Chat Page

Designed the AI Chat feature to support academic and career planning.

Allowed users to interact with the AI assistant after logging in.

Supported personalized planning for students' future development.

